

Лабораторная работа №8

Операционные системы

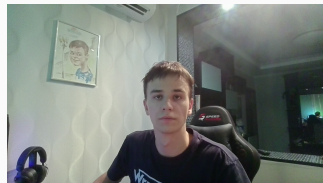
Савкин Д. В.

02 сентября 2026

Российский университет дружбы народов, Москва, Россия

Информация

- Савкин Дмитрий Вадимович
- РУДН, группа НПИбд-02-25
- Тема: поиск файлов, перенаправление, процессы



Цель работы

- Освоить поиск файлов, фильтрацию текста и управление процессами

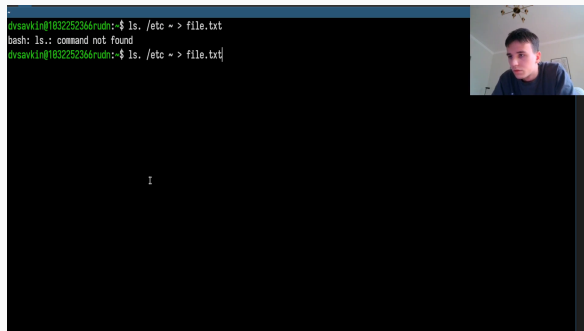
Задание

Задание

- Поиск и фильтрация файлов через find/grep
- Фоновые процессы, определение PID, df/du

Выполнение работы

Составляем список файлов /etc и домашнего каталога в file.txt.

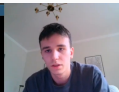


```
dvsavkin@1832252366rudn:~$ ls. /etc ~ > file.txt
bash: ls.: command not found
dvsavkin@1832252366rudn:~$ ls. /etc ~ > file.txt|
```

The image shows a terminal window with a dark background. The prompt is `dvsavkin@1832252366rudn:~$`. The user enters `ls. /etc ~ > file.txt`, which results in the error `bash: ls.: command not found`. The user then enters `ls. /etc ~ > file.txt|`. In the top right corner of the terminal window, there is a small video feed of a man with dark hair, wearing a blue shirt, looking towards the camera.

Выбираем .conf-файлы в conf.txt.

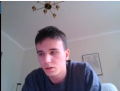
```
dvsavkin@1832252366rudn:~$ ls /etc ~ > file.txt
bash: ls.: command not found
dvsavkin@1832252366rudn:~$ ls /etc ~ > file.txt
dvsavkin@1832252366rudn:~$ grep .conf file.txt > conf.txt
bash: grep: command not found
dvsavkin@1832252366rudn:~$ grep .conf file.txt > conf.txt
dvsavkin@1832252366rudn:~$ |
```



Поиск файлов на “с”

Находим файлы, начинающиеся на “с”, несколькими способами.

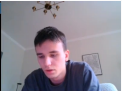
```
dv.savkin@1832252366rudn:~$ ls /etc ~ > file.txt
bash: ls.: command not found
dv.savkin@1832252366rudn:~$ ls /etc ~ > file.txt
dv.savkin@1832252366rudn:~$ grep .conf file.txt > conf.txt
bash: grep: command not found
dv.savkin@1832252366rudn:~$ grep .conf file.txt > conf.txt
dv.savkin@1832252366rudn:~$ find ~ -maxdepth 1 -name "с*"
dv.savkin@1832252366rudn:~$ ls | grep ^с
.conf.txt
dv.savkin@1832252366rudn:~$ find ~ -maxdepth 1 -name "с*"
/home/dv.savkin/.conf.txt
dv.savkin@1832252366rudn:~$ |
```



Постраничный вывод

Постранично выводим файлы /etc на букву "h" через less.

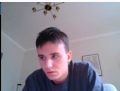
```
dv.savkin@1832252366rudn:~$ ls. /etc ~ > file.txt
bash: ls.: command not found
dv.savkin@1832252366rudn:~$ ls /etc ~ > file.txt
dv.savkin@1832252366rudn:~$ grep .conf file.txt > conf.txt
bash: grep: command not found
dv.savkin@1832252366rudn:~$ grep .conf file.txt > conf.txt
dv.savkin@1832252366rudn:~$ find ~ -maxdepth 1 -name "C*"
dv.savkin@1832252366rudn:~$ ls | grep ^c
.conf.txt
dv.savkin@1832252366rudn:~$ find ~ -maxdepth 1 -name "c*"
/home/dv.savkin/conf.txt
dv.savkin@1832252366rudn:~$ ls /etc | grep ^h | less
dv.savkin@1832252366rudn:~$ fi|
```



Фоновый процесс в logfile

Запускаем фоновый процесс, пишущий имена файлов на “log” в ~/logfile.

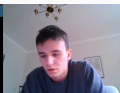
```
dv.savkin@1832252366rudn:~$ ls /etc ~ > file.txt
bash: ls.: command not found
dv.savkin@1832252366rudn:~$ ls /etc ~ > file.txt
dv.savkin@1832252366rudn:~$ grep .conf file.txt > conf.txt
bash: grep: command not found
dv.savkin@1832252366rudn:~$ grep .conf file.txt > conf.txt
dv.savkin@1832252366rudn:~$ find ~ -maxdepth 1 -name "C*"
dv.savkin@1832252366rudn:~$ ls | grep ^c
.conf.txt
dv.savkin@1832252366rudn:~$ find ~ -maxdepth 1 -name "c*"
/home/dv.savkin/.conf.txt
dv.savkin@1832252366rudn:~$ ls /etc | grep ^h | less
dv.savkin@1832252366rudn:~$ find / -iname "**log*" > ~/logfile 2>/dev/null &
```



Удаление logfile

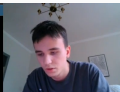
Удаляем ~/logfile.

```
dvsavkin@1832252366rudn:~$ ls /etc ~ > file.txt
bash: ls.: command not found
dvsavkin@1832252366rudn:~$ ls /etc ~ > file.txt
dvsavkin@1832252366rudn:~$ grep .conf file.txt > conf.txt
bash: grep: command not found
dvsavkin@1832252366rudn:~$ grep .conf file.txt > conf.txt
dvsavkin@1832252366rudn:~$ find ~ -maxdepth 1 -name "C*"
dvsavkin@1832252366rudn:~$ ls | grep ^c
.conf.txt
dvsavkin@1832252366rudn:~$ find ~ -maxdepth 1 -name "c*"
/home/dvsavkin/conf.txt
dvsavkin@1832252366rudn:~$ ls /etc | grep ^h | less
dvsavkin@1832252366rudn:~$ find / -iname "**log*" > ~/logfile 2>/dev/null &
[2] 24313
dvsavkin@1832252366rudn:~$ jobs
[1]-  Запущен          LIBGL_ALWAYS_SOFTWARE=1 firefox https://esystem.rudn.ru/course/view.php?id=5798 &
[2]+  Выход 1          find / -iname "**log*" > ~/logfile 2> /dev/null
dvsavkin@1832252366rudn:~$ rm ~/
```



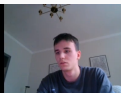
Запускаем gedit в фоне.

```
dvysavkin@1832252366rudn:~$ ls /etc ~ > file.txt
bash: ls.: command not found
dvysavkin@1832252366rudn:~$ ls /etc ~ > file.txt
dvysavkin@1832252366rudn:~$ grep .conf file.txt > conf.txt
bash: grep: command not found
dvysavkin@1832252366rudn:~$ grep .conf file.txt > conf.txt
dvysavkin@1832252366rudn:~$ find ~ -maxdepth 1 -name "C*"
dvysavkin@1832252366rudn:~$ ls | grep ^c
.conf.txt
dvysavkin@1832252366rudn:~$ find ~ -maxdepth 1 -name "c*"
/home/dvysavkin/conf.txt
dvysavkin@1832252366rudn:~$ ls /etc | grep ^h | less
dvysavkin@1832252366rudn:~$ find / -iname "**log*" > ~/logfile 2>/dev/null &
[2] 24313
dvysavkin@1832252366rudn:~$ jobs
[1]-  Запущен          LIBGL_ALWAYS_SOFTWARE=1 firefox https://esystem.rudn.ru/course/view.php?id=5798 &
[2]+  Выход 1          find / -iname "**log*" > ~/logfile 2>/dev/null
dvysavkin@1832252366rudn:~$ rm ~/logfile
dvysavkin@1832252366rudn:~$ Safe Browsing server returned a 400 during update:request url = https://safebrowsing.googleapis.com/v5/hashLists:batchGet?key=AIzaSyBPGKa4AYD4FC3HJK7LnIKxm4FDusVuucc, payload =
firefoq
```



Определяем PID gedit через ps+grep, вторым способом.

```
dvysavkin@1832252366rudn:~$ grep .conf file.txt > conf.txt
bash: grep: command not found
dvysavkin@1832252366rudn:~$ grep .conf file.txt > conf.txt
dvysavkin@1832252366rudn:~$ find ~ -maxdepth 1 -name "c*"
dvysavkin@1832252366rudn:~$ ls | grep ^c
.conf.txt
dvysavkin@1832252366rudn:~$ find ~ -maxdepth 1 -name "c*"
/home/dvysavkin/conf.txt
dvysavkin@1832252366rudn:~$ ls /etc | grep ^h | less
dvysavkin@1832252366rudn:~$ find / -iname "**log*" > ~/logfile 2>/dev/null &
[2] 24313
dvysavkin@1832252366rudn:~$ jobs
[1]-  Запущен          LIBGL_ALWAYS_SOFTWARE=1 firefox https://esystem.rudn.ru/course/view.php?id=5798 &
[2]+  Выход 1          find / -iname "**log*" > ~/logfile 2>/dev/null
dvysavkin@1832252366rudn:~$ rm ~/logfile
dvysavkin@1832252366rudn:~$ Safe Browsing server returned a 400 during update:request url = https://safebrowsing.googleapis.com/v5/hashLists:batchGet?key=AIzaSyBPGXa4AYD4FC3HJK7LnIKxm4FDusVuucc, payload =
firefox --new-instance &
[2] 27721
dvysavkin@1832252366rudn:~$
```



Завершение процесса

man kill, завершаем gedit.

```
dvsavkin 2856 0.0 0.8 454588 32492 pts/0 S1 17:15 0:00 /usr/lib64/firefox/firefox -contentpr
421110216 -sandboxingKind 0 -prefsHandle 0:45848 -prefMapHandle 1:285855 -sandboxReporter 2 -chrootClie
alChannelId {a68ff63b-5a8d-45e6-b9df-431978b0d0fb} -parentPid 2352 -crashReporter 5 -crashHelper 6 -appD
rowser 6 utility
dvsavkin 4078 0.1 1.9 2818232 75632 pts/0 S1 17:20 0:05 /usr/lib64/firefox/firefox -contentpr
Handle 0:42689 -prefMapHandle 1:285855 -jsInitHandle 2:156120 -parentBuildID 20260421110216 -sandboxRep
-ipcHandle 5 -initialChannelId {89232c06-fc4e-4f63-ba0f-9385a164297c} -parentPid 2352 -crashReporter 6 -crashHelper 7 -greomni
/usr/lib64/firefox/omni.ja -appomni /usr/lib64/firefox/browser/omni.ja -appDir /usr/lib64/firefox/browser 10 tab
dvsavkin 4085 0.1 1.9 2818232 75612 pts/0 S1 17:20 0:05 /usr/lib64/firefox/firefox -contentproc -isForBrowser -prefs
Handle 0:42689 -prefMapHandle 1:285855 -jsInitHandle 2:156120 -parentBuildID 20260421110216 -sandboxReporter 3 -chrootClient 4
-ipcHandle 5 -initialChannelId {1dc56cee-db65-4afc-88c8-a081f6419762} -parentPid 2352 -crashReporter 6 -crashHelper 7 -greomni
/usr/lib64/firefox/omni.ja -appomni /usr/lib64/firefox/browser/omni.ja -appDir /usr/lib64/firefox/browser 11 tab
dvsavkin 4185 0.0 1.9 2818232 75564 pts/0 S1 17:20 0:00 /usr/lib64/firefox/firefox -contentproc -isForBrowser -prefs
Handle 0:42689 -prefMapHandle 1:285855 -jsInitHandle 2:156120 -parentBuildID 20260421110216 -sandboxReporter 3 -chrootClient 4
-ipcHandle 5 -initialChannelId {96a15b3f-88bc-45fd-90da-7881db7625f} -parentPid 2352 -crashReporter 6 -crashHelper 7 -greomni
/usr/lib64/firefox/omni.ja -appomni /usr/lib64/firefox/browser/omni.ja -appDir /usr/lib64/firefox/browser 12 tab
dvsavkin 27721 5.3 3.4 3075472 137684 pts/0 S1 18:53 0:03 /usr/lib64/firefox/firefox --new-instance
dvsavkin 27738 0.0 0.0 149088 2792 ? S1 18:53 0:00 /usr/lib64/firefox/crashhelper 27721 9 /tmp/ 11
dvsavkin 27842 0.7 1.1 376956 47576 pts/0 S1 18:53 0:00 /usr/lib64/firefox/firefox -contentproc -parentBuildID 20260
421110216 -prefsHandle 0:25622 -prefMapHandle 1:283028 -sandboxReporter 2 -chrootClient 3 -ipcHandle 4 -initialChannelId {e5aaa
126-dc67-43f7-a3b8-1aef9a8e8daa} -parentPid 27721 -crashReporter 5 -crashHelper 6 -appDir /usr/lib64/firefox/browser 1 socket
dvsavkin 28080 0.0 0.0 231468 2796 pts/0 S+ 18:54 0:00 grep --color=auto firefox
dvsavkin@1832252366rudn:~$
```

Смотрим свободное место и объём каталога: df, du.

```

rowser 6 utility
dvsavkin 4078 0.1 1.9 2818232 75632 pts/0 S1 17:20 0:05 /usr/lib64/firefox/firefox -contentpr
Handle 0:42689 -prefMapHandle 1:285855 -jsInitHandle 2:156120 -parentBuildID 20260421110216 -sandboxRepc
-ipcHandle 5 -initialChannelId {09232c06-fc4e-4f63-ba0f-9385a164297c} -parentPid 2352 -crashReporter 6 -
/usr/lib64/firefox/omni.ja -appomni /usr/lib64/firefox/browser/omni.ja -appDir /usr/lib64/firefox/browse
dvsavkin 4085 0.1 1.9 2818232 75612 pts/0 S1 17:20 0:05 /usr/lib64/firefox/firefox -contentpr
Handle 0:42689 -prefMapHandle 1:285855 -jsInitHandle 2:156120 -parentBuildID 20260421110216 -sandboxReporter 3 -chrootClient 4
-ipcHandle 5 -initialChannelId {1dc56cee-db56-4afc-88c8-a081f6419762} -parentPid 2352 -crashReporter 6 -crashHelper 7 -greomni
/usr/lib64/firefox/omni.ja -appomni /usr/lib64/firefox/browser/omni.ja -appDir /usr/lib64/firefox/browser 11 tab
dvsavkin 4185 0.0 1.9 2818232 75564 pts/0 S1 17:20 0:00 /usr/lib64/firefox/firefox -contentproc -isForBrowser -prefs
Handle 0:42689 -prefMapHandle 1:285855 -jsInitHandle 2:156120 -parentBuildID 20260421110216 -sandboxReporter 3 -chrootClient 4
-ipcHandle 5 -initialChannelId {95a15b3f-88bc-45fd-90da-7881db7625f} -parentPid 2352 -crashReporter 6 -crashHelper 7 -greomni
/usr/lib64/firefox/omni.ja -appomni /usr/lib64/firefox/browser/omni.ja -appDir /usr/lib64/firefox/browser 12 tab
dvsavkin 27721 5.3 3.4 3075472 137684 pts/0 S1 18:53 0:03 /usr/lib64/firefox/firefox --new-instance
dvsavkin 27738 0.0 0.0 149088 2792 ? S1 18:53 0:00 /usr/lib64/firefox/crashhelper 27721 9 /tap/ 11
dvsavkin 27842 0.7 1.1 376956 47576 pts/0 S1 18:53 0:00 /usr/lib64/firefox/firefox -contentproc -parentBuildID 20260
421110216 -prefsHandle 0:25622 -prefMapHandle 1:283028 -sandboxReporter 2 -chrootClient 3 -ipcHandle 4 -initialChannelId {e5aaa
126-d6c7-43f7-a3b8-1aef9a8e8daa} -parentPid 27721 -crashReporter 5 -crashHelper 6 -appDir /usr/lib64/firefox/browser 1 socket
dvsavkin 28080 0.0 0.0 231468 2796 pts/0 S+ 18:54 0:00 grep --color=auto firefox
dvsavkin@1832252366rudn:~$ pgrep -x firefox
2352
27721
dvsavkin@1832252366rudn:~$ man

```

find директорий

Выводим все директории домашнего каталога через find.

```
dsavkin@183225236arubi:~$ find / -contatpr  
Handle 8:42689 -prefMapHandle 1:285855 -jsInitHandle 2:156128 -parentBuildID 28248421118216 -sandboxProc  
-jacHandle 5 -initialChannelID {89212036-1c4e-4f63-b0ff-9385a164291c} -parentPid 2252 -crashReporter 6 -  
/usr/lib64/firefox/omni-ja -appomi /usr/lib64/firefox/browser/omni-ja -appdir /usr/lib64/firefox/brows  
dsavkin 4885 0.1 1.9 2818232 15612 pts/0 SL 17:20 0:06 /usr/lib64/firefox/firefox -contatpr  
Handle 8:42689 -prefMapHandle 1:285855 -jsInitHandle 2:156128 -parentBuildID 28248421118216 -sandboxProc  
-jacHandle 5 -initialChannelID {1dc6c0e-dbf5-4af0-18d3-a801fa619762} -parentPid 2252 -crashReporter 6 -  
/usr/lib64/firefox/omni-ja -appomi /usr/lib64/firefox/browser/omni-ja -appdir /usr/lib64/firefox/brows  
dsavkin 4185 0.0 1.9 2818232 15644 pts/0 SL 17:20 0:00 /usr/lib64/firefox/firefox -contatpr -isforbrowser -pref  
Handle 8:42689 -prefMapHandle 1:285855 -jsInitHandle 2:156128 -parentBuildID 28248421118216 -sandboxReporter 3 -chromeClient 4  
-jacHandle 5 -initialChannelID {95a163f-88bc-45d0-96da-78811db7625f} -parentPid 2252 -crashReporter 6 -crashHalper 7 -groomi  
/usr/lib64/firefox/omni-ja -appomi /usr/lib64/firefox/browser/omni-ja -appdir /usr/lib64/firefox/brows  
dsavkin 27721 5.3 3.4 3875472 137884 pts/0 SL 18:53 0:03 /usr/lib64/firefox/firefox --new-instance  
dsavkin 27738 0.0 0.0 149886 2792 ? SL 18:53 0:00 /usr/lib64/firefox/crashhelper 27721 5 /tmp/11  
dsavkin 27842 0.7 1.1 376956 41576 pts/0 SL 18:53 0:00 /usr/lib64/firefox/firefox -contatpr -parentBuildID 28248  
42118216 -prefMapHandle 8:25622 -prefMapHandle 1:283828 -sandboxReporter 2 -chromeClient 3 -ipcHandle 4 -initialChannelID {c5aa  
124-d6c7-43f7-a3b0-1ae7a6b8daa} -parentPid 27721 -crashReporter 5 -crashHalper 6 -appdir /usr/lib64/firefox/brows  
dsavkin 28830 0.0 0.0 231468 2794 pts/0 S+ 18:54 0:00 grep --color=auto firefox  
dsavkin@183225236arubi:~$ pgrep -x firefox  
2252  
27721  
dsavkin@183225236arubi:~$ man kill  
dsavkin@183225236arubi:~$ kill 27721
```

```
/home/dsavkin/.gnupg  
/home/dsavkin/work  
/home/dsavkin/.qoro  
/home/dsavkin/lab2-report  
/home/dsavkin/lab3-report  
/home/dsavkin/git-extended  
/home/dsavkin/lab4-report  
/home/dsavkin/.password-store  
/home/dsavkin/bin  
/home/dsavkin/.bashrc.d  
/home/dsavkin/lab5-report  
/home/dsavkin/Downloads  
/home/dsavkin/Documents  
/home/dsavkin/lab6-report  
/home/dsavkin/ski_places  
/home/dsavkin/abc1  
/home/dsavkin/australia  
/home/dsavkin/play  
/home/dsavkin/lab7photos  
/home/dsavkin/lab7-report  
dsavkin@183225236arubi:~$ pgrep -x firefox  
2252  
dsavkin@183225236arubi:~$
```

```
dsavkin@183225236arubi:~$ man df  
dsavkin@183225236arubi:~$ df -h  
Filesystem      Размер  Использовано  Дост  Использованно  Смонтировано в  
/dev/sda3       76G      22G      53G      30% /  
devtmpfs        1.9G      0 1.9G      0% /dev  
tmpfs           1.9G      288K 1.9G      1% /dev/shm  
efivarfs        256K      215K 37K      86% /sys/firmware/efi/efivars  
tmpfs           777M      1.1M 776M      1% /run  
none            1.0M      0 1.0M      0% /run/credentials/systemd-journald.service  
none            1.0M      0 1.0M      0% /run/credentials/systemd-resolved.service  
tmpfs           1.9G      4.8K 1.9G      1% /tmp  
/dev/sda3       76G      22G      53G      30% /home  
/dev/sda2       2.0G      428M 1.4G      24% /boot  
/dev/sda1       599M      21M 579M      4% /boot/efi  
tmpfs           389M      76K 389M      1% /run/user/1000  
lab7photos     949G      466G 483G      50% /home/dsavkin/lab7photos  
lab7photos     949G      466G 483G      50% /program Files/Git/home/dsavkin/lab7photos  
lab7Photo      949G      466G 483G      50% /program Files/Git/home/dsavkin/lab7-report/qoro  
lab7Photo      949G      466G 483G      50% /home/dsavkin/lab7-report/adf184e8d8e5d1832c8b8e  
dsavkin@183225236arubi:~$ man du  
dsavkin@183225236arubi:~$ du -sh -  
613M  /home/dsavkin  
dsavkin@183225236arubi:~$
```

Выводы

- Освоены перенаправление ввода-вывода и поиск файлов
- Получены навыки управления фоновыми процессами и оценки места на диске